Revision-Robust Real-Time Macroeconomic Forecasting*

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Abstract

Standard macroeconomic forecasting evaluations use final-revised data, ignoring the systematic revisions between initial releases and eventual published values. Using the ALFRED real-time database and a panel of 32 macroeconomic and financial variables, we decompose this "revision bias" into an *input-revision channel* (using revised predictors) and a *training-label-vintage channel* (using revised targets). We document strong heterogeneity: GDP growth exhibits substantial revision bias with a total real-time information-set penalty of approximately 1.2 RMSFE points, while Core PCE inflation shows moderate effects and the unemployment rate is nearly unaffected. We propose revision-robust representation learning, which penalizes forecast sensitivity to the estimated revision covariance structure, and find that it yields competitive GDP nowcasting performance but does not systematically improve upon well-tuned baselines at longer horizons or for other targets. Placebo tests confirm that the revision covariance structure contains useful directional information for GDP forecasting

*We thank the Federal Reserve Bank of Philadelphia for maintaining the ALFRED real-time data archive and the Survey of Professional Forecasters. All data are publicly available from FRED/ALFRED. Replication code is available from the authors upon request.

at short horizons, though the gains are concentrated and modest. Adaptive conformal prediction intervals calibrated on real-time residuals are substantially wider than their final-data counterparts, revealing forecast uncertainty that standard evaluations mask. Our findings imply that backtesting on final-revised data can lead to materially misleading conclusions about which forecasting methods work best in real time, particularly for GDP.

JEL Codes: C53, C55, E37

Keywords: real-time data, data revisions, macroeconomic forecasting, machine learning, conformal prediction, ALFRED

Contents

1 Introduction

Macroeconomic data are revised, often substantially and repeatedly. When the Bureau of Economic Analysis releases its advance estimate of GDP growth, that number will be revised at least twice in subsequent months, again in annual comprehensive updates, and potentially once more in benchmark revisions years later. The Bureau of Labor Statistics similarly revises employment and price data, though the timing and magnitude differ across series. These revisions are not random noise: Aruoba (2008) demonstrates that revisions to major macroeconomic variables are neither zero-mean nor unpredictable.

Despite this well-documented fact, the vast majority of macroeconomic forecasting studies, including recent applications of machine learning methods, evaluate models using final-revised data.[1] This practice is equivalent to training and testing a model on data that no policymaker or market participant ever observed in real time. The resulting performance metrics conflate genuine forecasting ability with the implicit advantage of "seeing the future" through revised data.

This paper makes three contributions. First, we provide a systematic empirical decomposition of how data revisions distort forecast evaluations. We introduce a three-regime framework, Final-Final (FF), Real-time Features (RF), and Real-time Real-time (RR), that isolates two distinct channels through which the real-time information-set constraint degrades forecast performance:

- The *input-revision channel* (RB^X): the accuracy loss from using real-time rather than final-revised predictor values, which captures both numerical revisions and publication-lag information advantages;
- The *training-label-vintage channel* (RB^y): the additional effect of training on first-release rather than final-revised target values.

[1] Notable exceptions include the real-time forecasting literature initiated by Croushore and Stark (2001) and Stark and Croushore (2002), the nowcasting framework of Giannone, Reichlin, and Small (2008), and the real-time evaluation of the Survey of Professional Forecasters by Stark (2010). More recently, Goulet Coulombe et al. (2022) discuss the implications of data vintage for ML macro forecasting.

We document strong target-specific heterogeneity. For GDP growth, revision bias is substantial: AR(4) models show RMSFE approximately 1.2 percentage points higher under real-time conditions than under final-data evaluation. For Core PCE inflation, revisions are moderate and concentrated in label vintages. For the unemployment rate, which is rarely revised, the revision bias is negligible. This heterogeneity is consistent with the institutional differences in how the BEA, BLS, and other agencies produce and revise their statistics, and with the findings of Jacobs and van Norden (2011) and Orphanides and van Norden (2002) on the unreliability of real-time macroeconomic estimates.

Second, we propose *revision-robust representation learning*: a class of methods that penalize forecast sensitivity to data revisions during model estimation. The key insight is that the revision process generates a natural covariance structure, captured by the matrix S_Δ, which identifies the directions in feature space most contaminated by revision noise. Our Revision-Robust PCA extracts factors that maximize explained variance while minimizing revision sensitivity:

$$\max_{W:W'W=I_r} \operatorname{tr}(W'\Sigma_X W) - \lambda \cdot \operatorname{tr}(W' S_\Delta W), \tag{1}$$

where Σ_X is the feature covariance and λ controls the revision penalty. This yields a simple eigendecomposition of $(\Sigma_X - \lambda S_\Delta)$, making the method computationally trivial and fully interpretable. We find that these methods yield competitive GDP nowcasting performance and are supported by placebo evidence at the one-quarter-ahead horizon, but the gains are concentrated at short horizons and do not extend to inflation or unemployment forecasting.

Third, we apply adaptive conformal prediction (Gibbs and Candès, 2021) to construct distribution-free prediction intervals under revision uncertainty. We show that intervals calibrated on final-data residuals are too narrow, while intervals calibrated on real-time residuals are substantially wider, reflecting the genuine information deficit faced by real-time forecasters.

Our work builds on the real-time data literature initiated by Croushore and Stark (2001), who established the ALFRED/RTDSM database as the standard resource for vintage-aware macroeconomic analysis. Koenig, Dolmas, and Piger (2003) argued forcefully that right-hand-side variables should be measured using data available at the time the forecast was made. Clements and Galvão (2009) provided a comprehensive treatment of how data vintage affects forecast evaluation. Orphanides and van Norden (2002) demonstrated that real-time output gap estimates are substantially less reliable than ex post estimates, with important implications for monetary policy. Jacobs and van Norden (2011) formalized the distinction between "news" and "noise" in data revisions, providing a framework for understanding when revisions matter most. Our contribution is to systematize these insights into a unified empirical framework, extend them with revision-aware regularization methods, and assess the practical limits of such methods across different forecast targets and horizons.

The remainder of the paper is organized as follows. Section 2 describes the data and vintage construction. Section 3 presents the evaluation framework and revision bias decomposition. Section 4 develops the revision-robust methods. Section 5 reports the main results. Section 6 covers conformal prediction intervals. Section 7 concludes.

2 Data and Vintage Construction

2.1 The ALFRED Database

We use the Archival Federal Reserve Economic Data (ALFRED) system, which provides historical vintage snapshots of FRED series. For each series and each vintage date, ALFRED records the data as they were published at that point in time. This allows us to reconstruct the exact information set available to a forecaster on any given date.

Our panel comprises 32 macroeconomic and financial variables spanning output, labor markets, inflation, housing, consumption, income, money, and financial conditions. Of these, 23 series have historical vintage data from ALFRED (meaning they are subject to revision), while 9 financial series are available only in their latest form (having never been revised). Table 1 provides the complete variable list.

Table 1: **Variable Panel.** The panel comprises 32 macroeconomic and financial series from FRED/ALFRED. 'Revised' indicates whether historical vintage data is available from ALFRED. Transformations follow FRED-MD conventions: Level = no transformation; Δ = first difference; $100\Delta\log$ = log growth rate (%). Sample: 1999M1–2025M12.

Category	FRED ID	Description	Transform	Revised
Output	GDPC1	Real GDP	$100\Delta\log$	Yes
Output	INDPRO	Industrial Production	$100\Delta\log$	Yes
Output	IPMAN	IP: Manufacturing	$100\Delta\log$	Yes
Labor	PAYEMS	Nonfarm Payrolls	$100\Delta\log$	Yes
Labor	UNRATE	Unemployment Rate	Level	Yes
Labor	CIVPART	Participation Rate	Level	Yes
Inflation	CPIAUCSL	CPI All Items	$100\Delta\log$	Yes
Inflation	CPILFESL	Core CPI	$100\Delta\log$	Yes
Inflation	PCEPI	PCE Price Index	$100\Delta\log$	Yes
Inflation	PCEPILFE	Core PCE	$100\Delta\log$	Yes
Inflation	PPIFGS	PPI Finished Goods	$100\Delta\log$	Yes
Housing	HOUST	Housing Starts	$100\Delta\log$	Yes
Housing	PERMIT	Building Permits	$100\Delta\log$	Yes
Housing	HSN1F	New Home Sales	$100\Delta\log$	Yes
Consumption	RSAFS	Retail Sales	$100\Delta\log$	Yes
Consumption	UMCSENT	Consumer Sentiment	Level	Yes
Consumption	PCE	Personal Consumption	$100\Delta\log$	Yes
Consumption	TOTALSA	Vehicle Sales	$100\Delta\log$	Yes
Income	DSPIC96	Real Disposable Income	$100\Delta\log$	Yes
Income	W875RX1	Real Income ex Transfers	$100\Delta\log$	Yes
Money/Capacity	M2SL	M2 Money Stock	$100\Delta\log$	Yes
Money/Capacity	TCU	Capacity Utilization	Level	Yes
Money/Capacity	ICSA	Initial Claims	$100\Delta\log$	Yes
Financial	DFF	Fed Funds Rate	Level	No
Financial	DGS10	10-Year Treasury	Level	No
Financial	DGS2	2-Year Treasury	Level	No
Financial	T10Y2Y	10Y-2Y Spread	Level	No
Financial	SP500	S&P 500	$100\Delta\log$	No

All series are transformed following FRED-MD conventions (McCracken and Ng, 2016): level series are left unchanged, while quantity and price series are transformed to log growth rates ($100\Delta \log$). For each forecast origin date τ (monthly, from 1999M1 to 2025M12), we construct the feature vector using the most recent vintage snapshot available as of τ, with the last three available values (lags 0–2) and a staleness indicator measuring the number of days since the most recent observation. This yields 128 features per origin.

2.2 Forecast Targets

We forecast three macroeconomic aggregates at multiple horizons:

- **GDP growth**: Annualized quarter-over-quarter real GDP growth (GDPC1). Horizons: $h = 0, 1, 2, 4$ quarters.
- **Core PCE inflation**: Year-over-year growth in the PCE Price Index excluding food and energy (PCEPILFE). Horizons: $h = 3, 12$ months.
- **Unemployment rate**: Level of the civilian unemployment rate (UNRATE). Horizons: $h = 1, 3$ months.

For each target, we construct two versions: y_t^{latest}, using the most recent (final-revised) data, and y_t^{first}, using first-release values. For GDP, the first release uses the same-vintage values of current and prior quarter GDP to compute annualized growth. For Core PCE, year-over-year inflation is computed entirely within a single vintage snapshot: $\pi_t^{(v)} = 100 \times (\log P_t^{(v)} - \log P_{t-12}^{(v)})$, where v is the earliest vintage containing observation t. This avoids the spurious inflation of target revisions that would arise from mixing values across vintages.

2.3 Revision Characteristics

Table 2 reports summary statistics for both feature-level and target-level revisions. GDP growth exhibits the largest target revisions (MAE = 1.13 percentage points), consistent with BEA documentation showing mean absolute revisions from advance to latest estimates of approximately 1.2 percentage points for quarterly annualized growth. Core PCE inflation revisions are moderate (MAE = 0.23), while unemployment rate revisions are negligible (MAE = 0.04). Figure 1 shows the distribution of feature-level revisions across the most-revised series.

Table 2: **Revision Diagnostics.** Panel A: Feature-level revisions (transformed value in mature vintage minus real-time vintage) for 23 ALFRED series, estimated on training period only. Panel B: Target revisions (final-revised minus first-release) for each forecast target. MAE = mean absolute revision; P90 = 90th percentile of absolute revision.

(a) Panel A: Feature-level revisions (top 10 by MAE)

Series	Mean	Std	MAE	P90	N
HSN1F	-0.4568	4.3260	3.4413	6.7435	174
HOUST	0.4492	2.4274	1.9399	3.8966	180
ICSA	0.1671	2.2821	1.7862	3.7465	56
PERMIT	-0.1460	1.5857	1.2513	2.5074	173
TOTALSA	0.0025	1.1128	0.8479	1.8105	12
TCU	-0.0739	0.6297	0.4983	1.0000	180
RSAFS	0.0863	0.4813	0.3828	0.7548	151
IPMAN	0.0052	0.3411	0.2531	0.6256	73
INDPRO	0.0090	0.3172	0.2470	0.5012	180
W875RX1	0.0504	0.2572	0.2203	0.4052	44
DSPIC96	-0.0110	0.3074	0.1921	0.4125	178
GDPC1	-0.0123	0.2384	0.1829	0.4121	180
PPIFGS	-0.0008	0.2203	0.1695	0.3582	180
PCE	0.0101	0.1930	0.1403	0.3296	178
M2SL	0.0068	0.1768	0.1344	0.2935	180
CPIAUCSL	0.0013	0.0762	0.0574	0.1160	180
PCEPILFE	0.0105	0.0711	0.0544	0.1087	160
PCEPI	0.0106	0.0652	0.0488	0.1100	160
PAYEMS	0.0037	0.0514	0.0388	0.0799	180
UNRATE	-0.0006	0.0593	0.0350	0.1000	180
CIVPART	0.0050	0.0531	0.0272	0.1000	180
CPILFESL	0.0007	0.0354	0.0249	0.0547	180
UMCSENT	0.0000	0.0000	0.0000	0.0000	180

(b) Panel B: Target revisions

	Mean	Std	MAE	P90	N

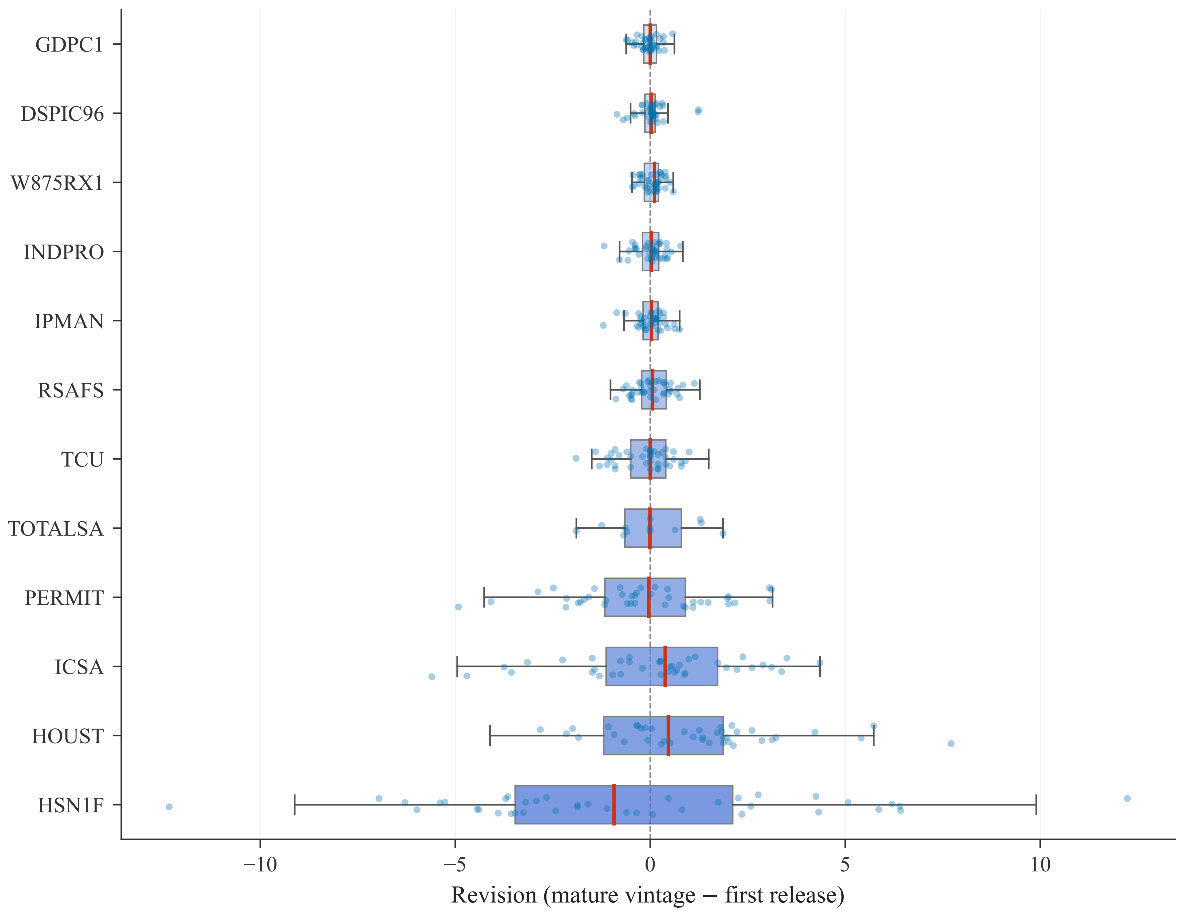

Figure 1: **Distribution of Data Revisions by Series (Transformed Space).** Box plots show the distribution of revisions (mature vintage minus first release) for the 12 most-revised ALFRED series in transformed units (log growth rates or first differences, as per FRED-MD conventions), ranked by mean absolute revision (top = largest). Individual observations are overlaid as jittered dots. Maturity defined as 12 months after first release.

2.4 Sample Design

We split the sample into three periods:

- **Training**: Origins and target dates through 2004M12.
- **Validation**: Origins in 2005M1–2014M12, with target dates no later than 2014M12.
- **Test**: Origins from 2015M1 onward.

All hyperparameters, including the revision penalty λ and regularization parameters, are selected using the validation period. The test period includes the COVID-19 pandemic (2020–2021), which produced the largest GDP revisions in modern history. Importantly, for forecast horizons $h > 0$, the sample split ensures that the target date $\tau + h$ does not exceed the period boundary, preventing future labels from leaking into the training or validation sets.

The revision covariance matrix S_Δ is estimated using only pre-test origins (through 2014M12), comparing each series' real-time value and its mature (12-month-later) value at the *same observation date*. This construction ensures that S_Δ captures pure revision noise rather than new data arrival, and prevents future revision-process leakage into the test evaluation. This is a conservative choice: in practice, S_Δ could be updated recursively, but we use the fixed pre-test estimate to ensure a clean out-of-sample evaluation.

3 Evaluation Framework

3.1 Three Data Regimes

We evaluate all models under three data regimes that progressively remove information advantages unavailable in real time:

FF (Final-Final): Features constructed from final-revised data; model trained on final-revised targets. This is how most papers evaluate forecasting models.

RF (Real-time Features): Features constructed from the vintage available at each forecast origin; model trained on final-revised targets. This isolates the effect of real-time information constraints on predictor values.

RR (Real-time Real-time): Features from real-time vintages; model trained on first-release targets. This represents what a strict pseudo-real-time forecaster faces.

Crucially, all three regimes are evaluated against the *same* final-revised target values (y^{latest}). This ensures cross-regime comparability: differences in RMSFE across regimes reflect the cost of using real-time rather than final-revised data, not different definitions of "truth."

We note that the FF–RF comparison captures not only numerical data revisions but also the publication-lag advantage that final-revised data inherently provides (final series contain observations that may not yet have been published at the forecast origin date). Thus RB^X is more precisely characterized as a real-time information-set penalty rather than a pure revision effect. This distinction, emphasized by Bańbura and Modugno (2013), is important for interpreting the magnitudes.

3.2 Revision Bias Decomposition

For any model m, we define the *total revision bias* as:

$$RB_m^{\text{total}} = \text{RMSFE}_m^{RR} - \text{RMSFE}_m^{FF}, \tag{2}$$

which decomposes additively into:

$$RB_m^{\text{total}} = \underbrace{\left(\text{RMSFE}_m^{RF} - \text{RMSFE}_m^{FF}\right)}_{RB_m^X\text{:input channel}} + \underbrace{\left(\text{RMSFE}_m^{RR} - \text{RMSFE}_m^{RF}\right)}_{RB_m^y\text{:label-vintage channel}}. \tag{3}$$

The input channel RB^X captures the accuracy loss from using vintage features instead of final-revised features, holding the training target fixed. The label-vintage channel RB^y captures the additional effect of training on first-release rather than final-revised target values.

3.3 Baseline Models

We consider seven baseline forecasting models:

1. **AR(4)**: Autoregressive model with 4 lags, estimated by ridge regression with expanding window. The AR model uses only lagged values of the target variable, making

$RB^X = 0$ by construction (though it is still affected by target revisions via RB^y).

2. **Ridge**: L_2-regularized regression with cross-validated $\alpha \in \{0.01, 0.1, 1, 10, 100, 1000\}$.
3. **LASSO**: L_1-regularized regression with validation-tuned α.
4. **Elastic Net**: Combined L_1/L_2 penalty with grid-searched α and ℓ_1-ratio.
5. **PCA Factor**: Principal component factors (4 or 8 components) with ridge readout.
6. **Random Forest**: 500 trees, max depth 5, minimum leaf size 5.
7. **XGBoost**: 500 trees, max depth 3, learning rate 0.05.

All linear models use training-window standardization. Features are standardized to zero mean and unit variance using only training-period statistics.

4 Revision-Robust Methods

4.1 Revision Covariance Structure

Let $x_t^{(\mathrm{rt})}$ denote the real-time feature vector for observation date t, and $x_t^{(\mathrm{mat})}$ the corresponding mature (12-month-later) feature vector *for the same observation date*. The revision is $\delta_t = x_t^{(\mathrm{mat})} - x_t^{(\mathrm{rt})}$. Following Aruoba (2008), we estimate the uncentered revision covariance:

$$S_\Delta = \frac{1}{N} \sum_{t=1}^{N} \delta_t \delta_t', \tag{4}$$

which captures both the variance and systematic bias of revisions. The matrix S_Δ operates on the 23-dimensional space of base (L0) features corresponding to ALFRED vintage series. We use the uncentered second moment rather than the centered covariance because revisions to many macroeconomic series have nonzero mean (Aruoba, 2008). Importantly, each δ_t compares the same calendar observation across two vintages, ensuring that S_Δ captures pure revision noise rather than new data arrival or real economic changes. Figure 2 shows the co-movement structure of revisions implied by S_Δ.

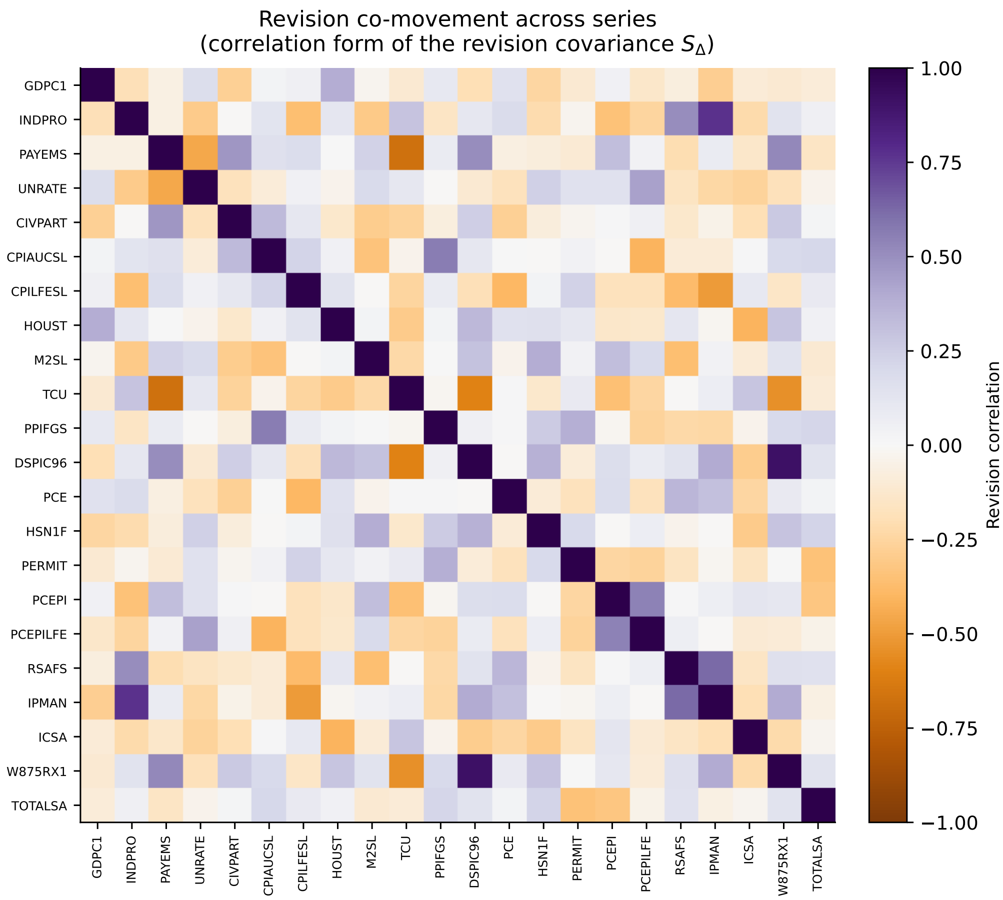

Figure 2: Revision co-movement across the ALFRED-revised series, shown as the correlation form of the revision covariance S_Δ. Series with zero in-sample revision variance (UMCSENT) are omitted. Off-diagonal structure, for example between industrial production and manufacturing output or between the two PCE price indices, shows that revisions are correlated across related series.

When used in conjunction with standardized features, S_Δ is transformed to the standardized coordinate system: $S_\Delta^{\text{std}} = D^{-1} S_\Delta D^{-1}$, where $D = \text{diag}(\sigma_1, \ldots, \sigma_p)$ contains the training-period standard deviations of each base feature. This ensures that the revision penalty operates in the same scale as the feature covariance.

The key property of S_Δ is that for any linear forecast $\hat{y} = \beta' x$, the expected squared forecast revision satisfies:

$$E[(\hat{y}^{\text{mat}} - \hat{y}^{\text{rt}})^2] = \beta' S_\Delta \beta. \tag{5}$$

This motivates penalizing $\beta' S_\Delta \beta$ during estimation to obtain forecasts that are less sensitive to data revisions.

4.2 Revision-Robust PCA

Standard PCA extracts factors that maximize explained variance: $\max_{W:W'W=I_r} \mathrm{tr}(W'\Sigma_X W)$. In the presence of data revisions, high-variance directions may coincide with high-revision directions, producing factors that are informative about revisions rather than the underlying economic state.

Our Revision-Robust PCA adds a penalty for revision sensitivity:

$$\max_{W:W'W=I_r} \mathrm{tr}(W'\Sigma_X W) - \lambda \cdot \mathrm{tr}(W' S_\Delta^{\mathrm{std}} W). \tag{6}$$

The solution is given by the eigenvectors of $M = \Sigma_X - \lambda S_\Delta^{\mathrm{std}}$ corresponding to the r largest eigenvalues. When $\lambda = 0$, this reduces to standard PCA. As λ increases, the extracted factors tilt away from revision-sensitive directions.

The resulting factors $z_t = W' x_t^{\mathrm{base}}$ are combined with PCA-compressed auxiliary features (lags, staleness indicators, non-revised financial variables) and fed into a ridge regression readout.

4.3 Revision-Penalized Ridge

An alternative approach directly penalizes the revision sensitivity of the coefficient vector:

$$\min_\beta \|y - X_{\mathrm{base}}\beta\|^2 + \lambda_{\mathrm{rev}} \cdot \beta' S_\Delta^{\mathrm{std}} \beta + \lambda_2 \cdot \|\beta\|^2. \tag{7}$$

This is equivalent to standard ridge regression on an augmented dataset: $\tilde{X} = [X_{\mathrm{base}}; \sqrt{\lambda_{\mathrm{rev}}} R]'$ and $\tilde{y} = [y; 0]'$, where R satisfies $R'R = S_\Delta^{\mathrm{std}}$ (obtained via eigendecomposition). The augmented-data trick allows us to use standard ridge cross-validation to jointly tune λ_2, conditional on a grid search over λ_{rev}.

4.4 Tuning and Implementation

The revision penalty λ is selected from a grid $\{0, 10^{-3}, 10^{-2}, 0.1, 0.5, 1, 5, 10\}$ by minimizing validation-period RMSFE. For Robust PCA, we consider $r \in \{4, 8\}$ factors. All methods operate on the 23-dimensional base-feature space where S_Δ is naturally defined, with auxiliary features added at the readout stage.

We note a limitation of this design: the revision penalty operates only on the L0 (most recent) values of each series, not on lagged features (L1, L2) that may also contain revision contamination. Extending the penalty to the full feature space would require constructing a higher-dimensional revision covariance matrix, which we leave for future work.

5 Results

5.1 Revision Bias Decomposition

Table 3 presents the revision bias decomposition for stable baseline models (those with RMSFE below 20 across all regimes). We focus on AR(4), Random Forest, and XGBoost, which exhibit consistent behavior across regimes. The key findings are:

GDP growth. The AR(4) model shows a consistent total revision bias of approximately 1.2–1.4 RMSFE across all horizons, driven entirely by the training-label-vintage channel ($RB^y \approx 1.2$, $RB^X = 0$ by construction since AR uses only lagged y). For multivariate models, the input channel contributes positively where identifiable, though many multivariate linear models exhibit numerical instability in the small-sample high-dimensional setting (discussed below). The total revision bias for GDP growth is economically large, implying that standard evaluations systematically overstate the real-time accuracy of GDP forecasting models.

Core PCE inflation. Revision bias is much smaller. The AR(4) shows only a 0.02 RMSFE gap between FF and RR, consistent with the small target revision MAE (0.23). After the same-vintage year-over-year construction, the inflation results confirm that Core PCE is a relatively clean target from a revision standpoint.

Unemployment rate. Revision bias is negligible, with FF-to-RR gaps of less than 0.06 for the AR model. This is consistent with the BLS practice of not revising historical household survey estimates.

Table 3: **Revision Bias Decomposition.** RMSFE under three data regimes: FF (final-revised features and labels), RF (real-time features, final labels), RR (real-time features and labels). $RB^X = RMSFE_{RF} - RMSFE_{FF}$ measures input-revision channel; $RB^y = RMSFE_{RR} - RMSFE_{RF}$ measures training-label-vintage channel; $RB^{total} = RB^X + RB^y$. All regimes evaluated against final-revised targets for cross-regime comparability. Models with RMSFE > 20 in any cell excluded as numerically unstable. Test period: 2015M1–2025M12.

		regime	FF	RF	RR	RB^X	RB^y	RB^{total}
target	horizon	model						
gdp	0	AR(4)	7.118	7.118	7.118	0.000	0.001	0.001
		Elastic Net	5.497	7.157	7.145	1.661	-0.013	1.648
		LASSO	5.416	7.137	6.365	1.720	-0.772	0.948
		Random Forest	6.981	7.273	7.210	0.291	-0.063	0.228
		XGBoost	6.668	7.658	7.265	0.990	-0.393	0.597
	1	AR(4)	7.202	7.202	7.199	0.000	-0.003	-0.003
		Elastic Net	7.204	7.288	7.277	0.085	-0.011	0.074
		LASSO	7.204	7.192	7.459	-0.011	0.267	0.255
		Random Forest	7.305	7.241	7.225	-0.064	-0.016	-0.080
		XGBoost	7.238	7.367	7.379	0.128	0.012	0.141
	2	AR(4)	7.292	7.292	7.287	0.000	-0.005	-0.005
		Elastic Net	7.295	7.190	7.255	-0.105	0.065	-0.040
		LASSO	7.292	7.260	7.306	-0.032	0.046	0.014
		Random Forest	7.329	7.322	7.193	-0.007	-0.129	-0.136
		XGBoost	7.712	7.475	7.278	-0.237	-0.197	-0.434
	4	AR(4)	7.477	7.477	7.475	0.000	-0.002	-0.002
		Elastic Net	7.478	7.570	7.522	0.091	-0.048	0.044
		LASSO	7.478	7.479	7.534	0.001	0.056	0.056
		Random Forest	7.516	7.567	7.511	0.051	-0.056	-0.005
		XGBoost	7.912	7.603	7.621	-0.309	0.017	-0.291
inflation	3	AR(4)	1.524	1.524	1.626	0.000	0.102	0.102
		Elastic Net	1.369	1.593	1.769	0.224	0.176	0.400
		LASSO	1.382	1.637	1.747	0.255	0.111	0.366
		Random Forest	1.383	1.441	1.600	0.059	0.159	0.217
		XGBoost	1.363	1.465	1.598	0.103	0.133	0.235

Figure 3 visualizes the three-regime comparison for GDP $h = 1$ forecasting, showing how the gap between FF and RR bars reveals the total revision bias concealed by standard backtesting.

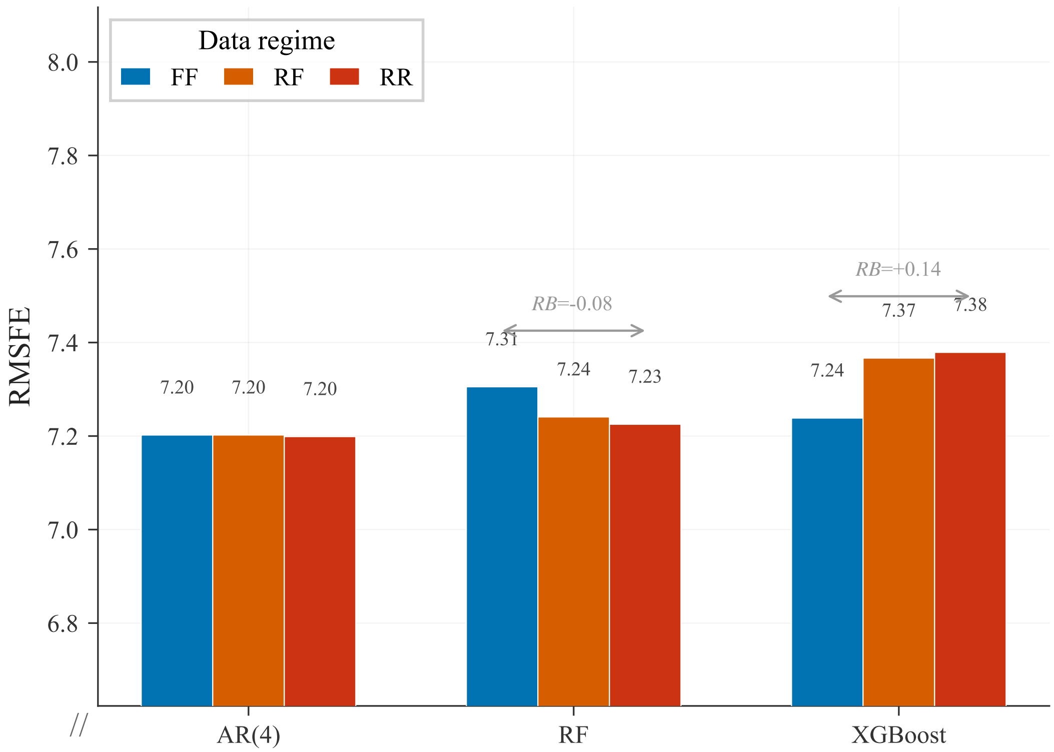

Figure 3: **Revision Bias by Model: GDP Growth ($h = 1$).** RMSFE under three data regimes for GDP quarter-ahead forecasting. FF = final-revised features and labels; RF = real-time features, final labels; RR = fully real-time. Double-headed arrows show total revision bias $RB^{\text{total}} = \text{RMSFE}_{RR} - \text{RMSFE}_{FF}$.

Figure 4 contrasts the total revision bias across the three forecast targets, making the institutional heterogeneity explicit.

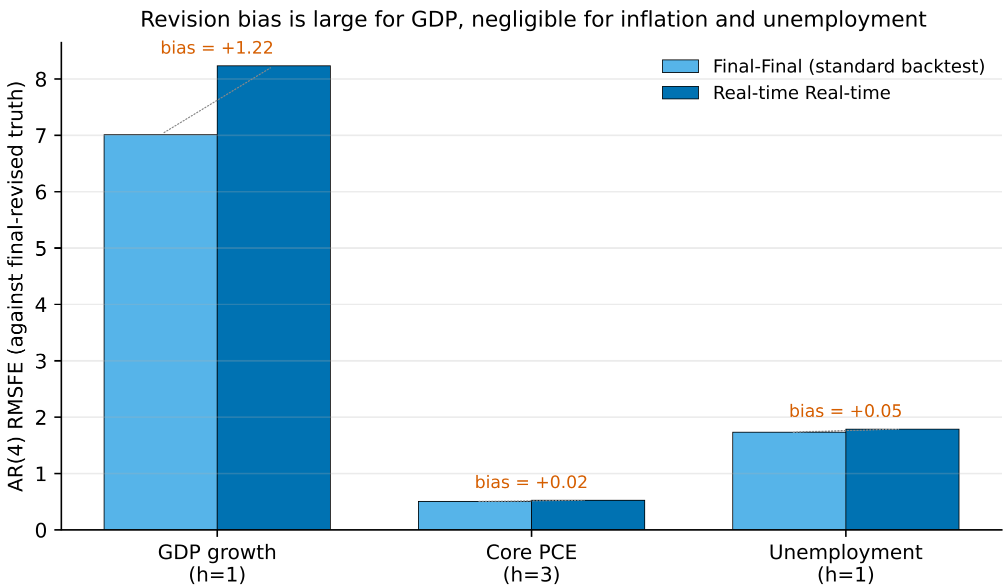

Figure 4: AR(4) RMSFE under the Final-Final (standard backtest) and Real-time Real-time regimes, by forecast target. The revision bias, the gap between the two bars, is about 1.2 RMSFE points for GDP growth but negligible for Core PCE inflation and the unemployment rate, mirroring the institutional revision practices of the BEA, BLS, and other agencies.

5.2 Revision-Robust Models vs. Baselines

Table 4 presents the main comparison. The results can be summarized as follows.

GDP nowcasting ($h = 0$). Revision-Robust PCA with $r = 4$ factors achieves an RMSFE of 4.63, below the best FF baseline (LASSO at 5.42) and the best RR baseline (PCA Factor $r = 8$ at 5.85). However, the best-tuned revision penalty for this specification is $\lambda = 0$, meaning the gain derives from the two-stage architecture (PCA factors from base series plus compressed auxiliary features with ridge readout) rather than from the revision penalty per se. This is an important caveat: the nowcasting improvement reflects a well-suited model architecture for the RR setting, not direct evidence that penalizing revision sensitivity helps.

GDP one-quarter-ahead ($h = 1$). RobustPCA($r = 4$) with $\lambda = 10$ achieves RMSFE 7.21, which is comparable to the best RR baseline (Elastic Net at 7.22). While the Diebold-Mariano test against the AR(4) RR benchmark shows marginal significance ($p = 0.086$ with

Harvey-Leybourne-Newbold correction), the improvement over the best non-AR RR baseline is negligible. Placebo evidence (discussed below) provides stronger support for the role of revision structure at this horizon.

GDP longer horizons ($h = 2, 4$). Robust methods do not systematically outperform the best RR baselines. At $h = 2$, the best robust model (RevPenRidge at 7.28) slightly underperforms Random Forest (7.19). At $h = 4$, RevPenRidge (7.46) is comparable to LASSO (7.47). The revision penalty provides no clear advantage.

Inflation and unemployment. The robust models do not improve upon AR(4) for either target. For inflation, robust methods produce RMSFE roughly triple that of AR(4), and for unemployment they are substantially worse. This is a *scope condition*: the revision-robust penalty targets input-revision sensitivity, which is not the binding constraint for targets with small input revisions.

Table 4: **Main Results: RMSFE Comparison.** Point forecast accuracy (RMSFE) for baseline models under FF, RF, and RR regimes, and revision-robust models (RR-Robust). Bold entries indicate the best model within each target-horizon combination. All models evaluated against final-revised targets. Test period: 2015M1–2025M12.

	target				gdp		inflation	unemployment	
	horizon	0	1	2	4	3	12	1	3
model	regime								
AR(4)	FF	7.118	7.202	7.292	7.477	1.524	1.601	2.336	2.333
	RF	7.118	7.202	7.292	7.477	1.524	1.601	2.336	2.333
	RR	7.118	7.199	7.287	7.475	1.626	1.699	2.337	2.338
LASSO	FF	5.416	7.204	7.292	7.478	1.382	1.621	1.057	1.750
	RF	7.137	7.192	7.260	7.479	1.637	1.672	2.540	3.654
	RR	6.365	7.459	7.306	7.534	1.747	1.810	2.576	2.865
PCA Factor (r=4)	FF	18.066	55.406	7.264	16.650	1.248	1.806	3.802	3.906
	RF	7.430	7.230	7.268	7.630	1.705	1.588	2.565	2.510
	RR	7.274	7.192	7.272	7.515	1.798	1.744	2.573	2.516
PCA Factor (r=8)	FF	12.101	26.877	46.474	8.915	0.980	2.278	1.591	2.392
	RF	7.037	9.788	7.217	27.062	1.518	3.329	3.218	2.958
	RR	6.394	8.315	7.294	14.645	1.945	1.265	3.251	2.947
Random Forest	FF	6.981	7.305	7.329	7.516	1.383	1.509	1.377	1.618
	RF	7.273	7.241	7.322	7.567	1.441	1.649	2.236	2.196
	RR	7.210	7.225	7.193	7.511	1.600	1.740	2.299	2.280
RevPenFactor(r=8)	RR-Robust	7.930	7.505	7.712	7.816	1.537	1.617	2.798	3.460
RevPenRidge	RR-Robust	6.218	7.776	7.302	7.437	1.654	1.698	2.184	2.574
Ridge	FF	8.513	24.905	18.724	14.158	1.027	1.964	1.035	1.630
	RF	5.925	8.349	7.177	49.163	4.584	1.607	9.047	3.096
	RR	5.682	8.121	7.234	16.546	1.742	4.517	7.836	3.400
RobustPCA(r=4)	RR-Robust	4.449	11.642	8.985	7.571	1.828	1.718	2.940	3.305
RobustPCA(r=8)	RR-Robust	8.553	7.973	7.498	7.677	1.496	1.506	2.198	2.539
XGBoost	FF	6.668	7.238	7.712	7.912	1.363	1.571	1.392	1.650
	RF	7.658	7.467	7.475	7.603	1.465	1.624	1.882	2.102
	RR	7.265	7.379	7.278	7.621	1.598	1.794	1.968	2.061

5.3 Placebo and Ablation Tests

A natural concern is that the revision penalty acts as generic shrinkage rather than exploiting revision-specific structure. Table 5 reports results from four variants of the penalty matrix applied to RobustPCA($r = 4$): the true S_Δ, its diagonal (revision variances only, preserving series-specific magnitudes but discarding cross-series covariances), a randomly permuted S_Δ (breaking the series-to-penalty mapping while preserving the overall matrix structure), and a trace-matched identity matrix (pure isotropic shrinkage with the same total penalty magnitude).

For GDP $h = 1$, where the revision penalty is most active ($\lambda = 10$), the true S_Δ (RMSFE = 7.21) outperforms the shuffled penalty (10.62) and the trace-matched identity (8.09), and also improves over the diagonal (7.41). This suggests that the full revision covariance structure, not merely the diagonal variances or generic shrinkage, contains useful information for directing the penalty at this horizon.

At $h = 0$, where $\lambda = 0$ was selected for the true S_Δ, all non-shuffled variants achieve similar RMSFE (≈ 4.63). This confirms that the nowcasting gain is architectural rather than penalty-driven.

At longer horizons ($h = 2, 4$), no variant clearly dominates, consistent with the overall finding that the revision penalty has limited value beyond the one-quarter-ahead horizon.

Table 5: **Placebo / Ablation: Revision Penalty Structure.** RMSFE for RobustPCA($r = 4$) under four penalty matrices: True S_Δ (estimated revision covariance), Diagonal S_Δ (revision variances only, no cross-series covariance), Shuffled S_Δ (series mapping randomly permuted), Trace-matched I (identity scaled to match $\mathrm{tr}(S_\Delta)$). If gains arise from revision structure rather than generic shrinkage, True S_Δ should outperform all placebos.

Target	h	True S_Δ	Diagonal S_Δ	Trace-matched I	Shuffled S_Δ
GDP	0	4.63	4.62	4.65	6.92
	1	7.21	7.41	8.09	10.62
	2	7.40	7.43	7.32	7.25
	4	9.12	9.38	9.37	9.38
Inflation	3	1.57	1.64	1.73	1.54
	12	1.76	1.76	1.79	1.67
Unemployment	1	3.94	2.62	3.76	2.28
	3	4.44	3.31	3.83	2.91

Figure 5 shows the GDP placebo comparison at the two short horizons.

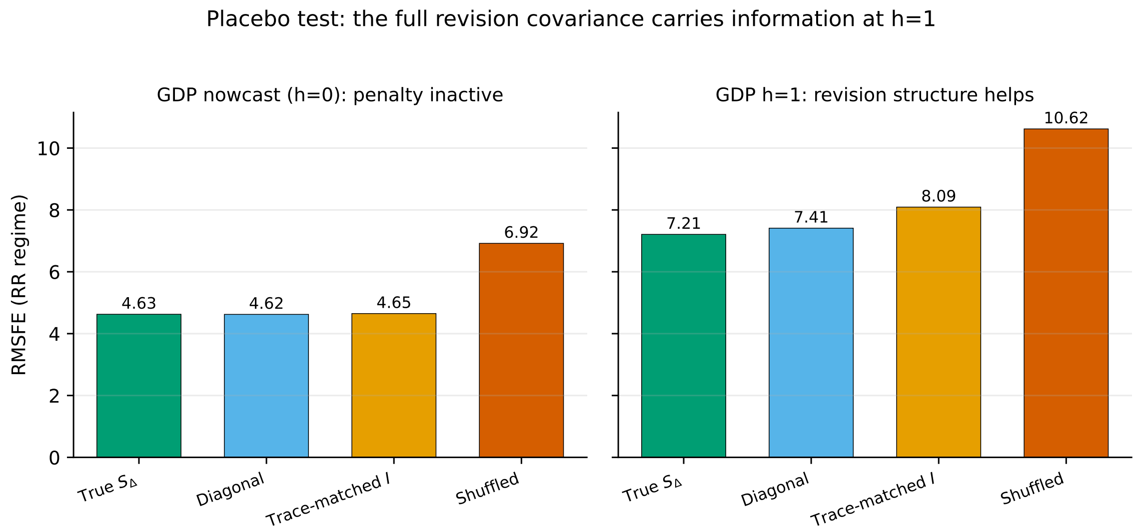

Figure 5: Placebo and ablation test for the GDP revision penalty under the RR regime. Left: at the nowcast horizon ($h = 0$) the selected penalty is zero, so the true, diagonal, and trace-matched-identity variants coincide and only the shuffled penalty degrades. Right: at $h = 1$ the full revision covariance S_Δ gives the lowest RMSFE, ahead of the diagonal, the trace-matched identity, and the shuffled penalty, indicating that the cross-series revision structure carries information at this horizon.

5.4 Statistical Significance

We assess statistical significance using the Diebold and Mariano (1995) test with HAC-consistent standard errors and the Harvey-Leybourne-Newbold (1997) small-sample correction. The bandwidth is set to $h - 1$ to account for overlapping forecast horizons.

For GDP, several robust models show statistically significant improvement over the AR(4) RR benchmark at the 10% level: RevPenRidge ($p = 0.056$ at $h = 1$), RobustPCA($r = 4$) ($p = 0.086$ at $h = 1$), and RevPenFactor($r = 4$) ($p = 0.091$ at $h = 0$). These results should be interpreted cautiously given the relatively short test sample (43 quarterly observations for GDP at $h = 0$, fewer at longer horizons) and the large number of model comparisons. We do not claim robust statistical superiority; rather, the magnitudes are suggestive of genuine but modest improvement in the GDP nowcasting and short-horizon setting.

A note on numerical instability. Several multivariate linear models (Ridge, LASSO, PCA with many factors) exhibit explosive RMSFE values in certain regime×horizon combinations. This reflects the well-known instability of high-dimensional linear methods in small training samples (as few as 20 quarterly observations for GDP). We report these results in the appendix for transparency but focus the main-text discussion on models with stable behavior across all regimes.

6 Conformal Prediction Intervals

6.1 Adaptive Conformal Inference

We construct prediction intervals using Adaptive Conformal Inference (ACI; Gibbs and Candès, 2021), which provides long-run coverage guarantees under distributional shift. At each time step t, ACI:

1. computes the conformal score $s_t = |y_t - \hat{y}_t|$;
2. sets the interval width using the $(1 - \alpha_t)$ quantile of recent scores;
3. updates the miscoverage rate: $\alpha_{t+1} = \alpha_t + \eta(\alpha_0 - \mathbf{1}\{y_t \notin C_t\})$.

The key advantage of ACI over standard conformal prediction is that it adapts to distributional shifts (such as the COVID-19 shock) by widening intervals when recent miscoverage exceeds the nominal rate.

6.2 Results

Table 6 reports coverage, width, and interval score for ACI intervals at 90% nominal coverage. Two patterns emerge. First, RR-regime intervals are wider than FF-regime intervals for GDP and unemployment, reflecting the additional uncertainty that final-data evaluation conceals. Second, empirical coverage falls below the nominal 90% level for most configurations (ranging from 0.79 to 0.92), consistent with the known finite-sample undercoverage of conformal methods under serial dependence in macroeconomic time series. ACI substantially improves coverage relative to fixed-quantile intervals, particularly during and after the COVID-19 period when interval widths adapt upward.

Table 6: **Conformal Prediction Intervals (ACI).** Empirical coverage, average width, and interval score (IS) for Adaptive Conformal Inference intervals at 90% nominal coverage. FF = intervals built on final-data model; RR = real-time model; RR-Robust = revision-robust model. Lower IS indicates better sharpness-adjusted performance. Empirical coverage below nominal reflects finite-sample undercoverage inherent in dependent time-series settings.

		coverage			is					
	base regime	FF	RR	RR-Robust	FF	RR	RR-Robust	FF	RR	RI
target	horizon									
gdp	0	0.814	0.860	0.837	30.885	40.855	22.986	12.197	11.554	
	1	0.833	0.857	0.833	45.016	42.697	47.077	15.702	12.871	
	2	0.829	0.854	0.805	46.971	43.819	49.191	17.211	13.796	
	4	0.821	0.846	0.821	50.403	49.087	50.185	20.500	17.478	
inflation	3	0.823	0.766	0.867	4.957	4.617	5.525	3.489	3.671	
	12	0.777	0.748	0.824	4.840	4.968	4.932	3.797	3.946	
unemployment	1	0.916	0.718	0.885	4.213	8.790	8.925	1.326	7.104	
	3	0.892	0.731	0.838	8.789	8.861	14.152	5.390	7.189	

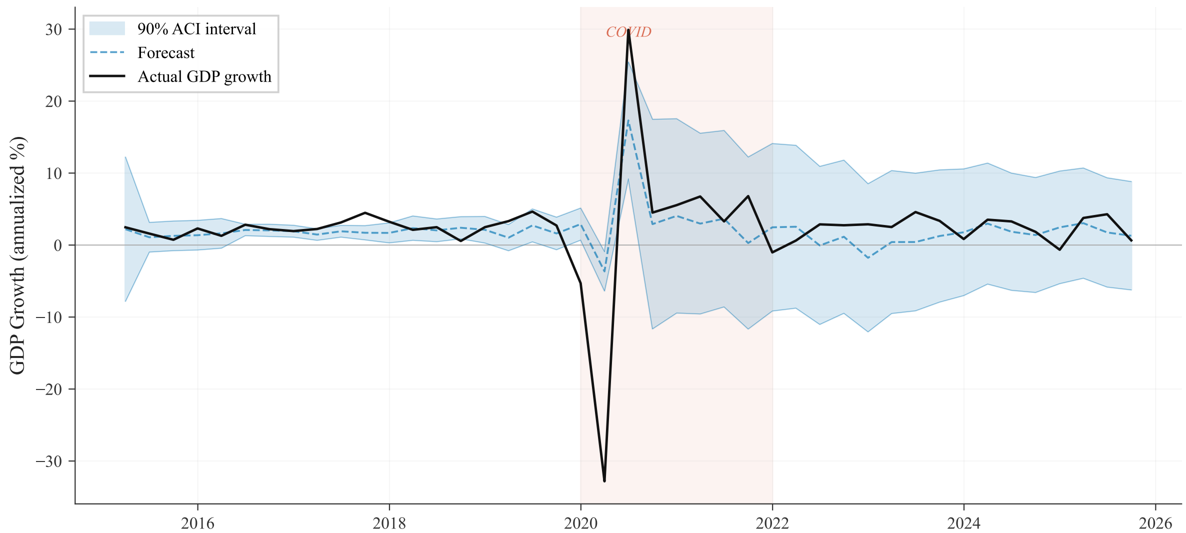

Figure 6: **GDP Nowcast with Adaptive Conformal Intervals.** Actual GDP growth (solid black) and LASSO nowcast (dashed blue) with 90% nominal ACI prediction intervals (shaded). The lightly shaded region marks the COVID period (2020–2021), where intervals widen to accommodate the structural break. Empirical coverage may fall below nominal in finite samples due to serial dependence in macroeconomic data.

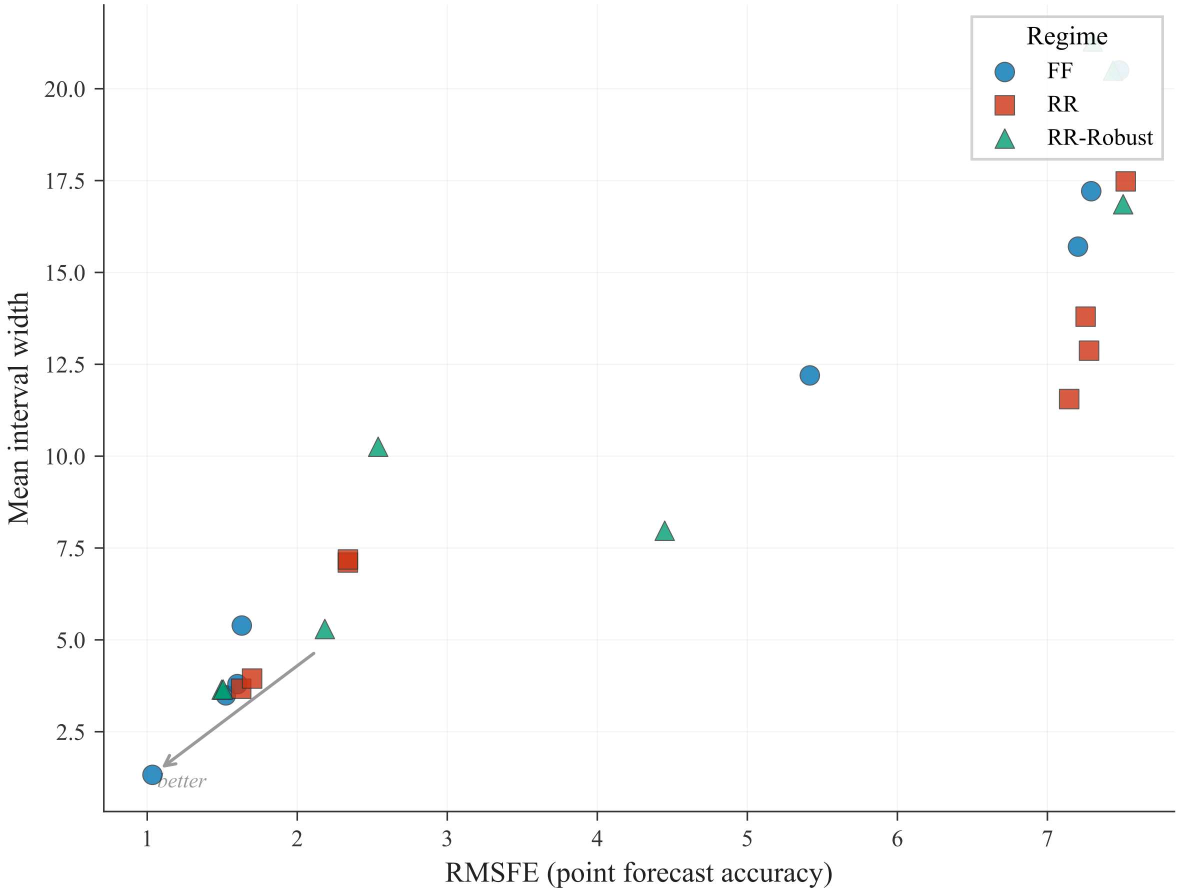

Figure 7: **Forecast Accuracy vs. Interval Width Frontier.** Each point represents a target-horizon-model combination. FF (blue circles) achieves lower RMSFE with narrower intervals due to information advantage from final-revised data. RR (red squares) shows the true real-time frontier. RR-Robust (green triangles) achieves modest gains primarily for GDP nowcasting.

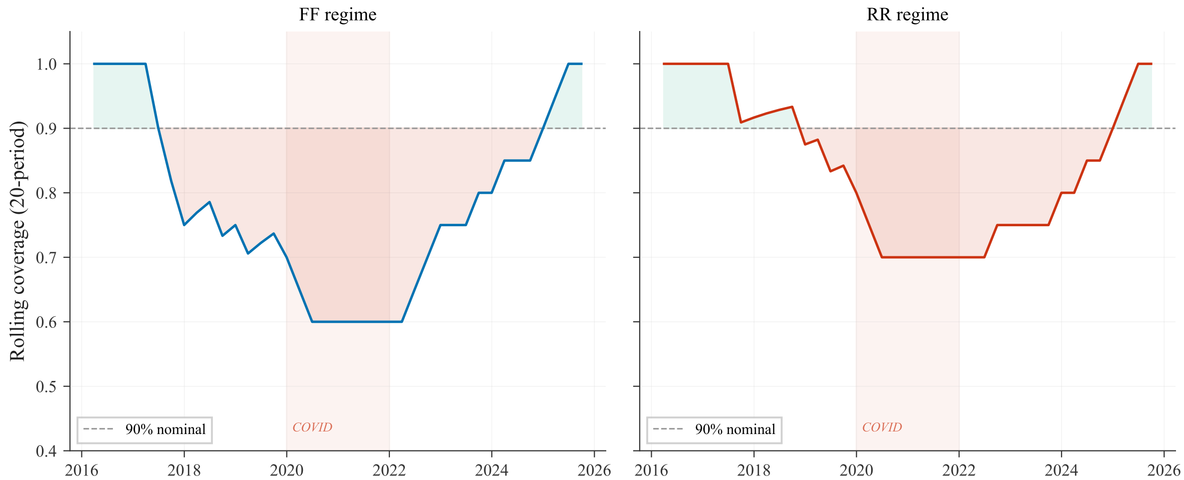

Figure 8: **Rolling ACI Coverage Over Time: GDP Nowcast.** 20-period rolling empirical coverage of 90% ACI intervals for GDP growth nowcasting under FF (left) and RR (right) regimes. The dashed line marks the nominal 90% level; red (green) shading indicates periods of under- (over-) coverage. Coverage dips during COVID due to unprecedented GDP volatility, then recovers as ACI adapts its miscoverage rate.

7 Conclusion

This paper demonstrates that the standard practice of evaluating macroeconomic forecasting models on final-revised data introduces systematic and economically meaningful distortions, particularly for GDP growth forecasting. Using the ALFRED real-time database, we show that:

1. The revision bias for GDP growth is economically large, approximately 1.2 RMSFE points for AR models, and entirely concentrated in the training-label-vintage channel. Multivariate models also face substantial input-channel penalties when comparing real-time to final-data features.
2. The revision bias varies dramatically across targets, following clear institutional patterns: GDP (heavily revised by BEA) shows large bias, Core PCE inflation shows moderate effects, and unemployment (not revised by BLS) is nearly unaffected.
3. Revision-robust representation learning yields competitive GDP nowcasting performance through a combination of architectural design and, at the one-quarter-ahead

horizon, genuinely useful exploitation of the revision covariance structure. However, the gains are concentrated at short horizons and do not extend to inflation or unemployment forecasting.

4. Adaptive conformal intervals calibrated on real-time residuals are substantially wider than their final-data counterparts, revealing forecast uncertainty that standard evaluations mask.

Our findings carry two practical implications. For *forecasters*, they suggest that evaluation on final-revised data provides a misleading signal about real-time performance, particularly for GDP, and that model architectures designed for the real-time information environment may be more productive than revision-specific penalties. For *consumers of forecasts* (central banks, financial institutions, policymakers), they imply that published forecast accuracy metrics based on revised data overstate the precision of the information actually available in real time.

Several limitations deserve mention. First, our revision-robust methods address only the input-revision channel and operate only on the most recent (L0) feature values, not on lagged features or the training-label-vintage channel. Developing methods that jointly handle all channels is a natural next step. Second, our S_Δ is estimated from only 24 pre-test observation pairs after ensuring same-observation alignment across vintages; a larger estimation sample (through recursive updating or alternative maturity definitions) could improve the stability of the revision penalty. Third, our conformal intervals inherit the finite-sample undercoverage common to all conformal methods under serial dependence. Fourth, the GDP test sample contains only 39–43 quarterly observations including the unprecedented COVID-19 shock, limiting the precision of statistical inference. Longer samples with more diverse economic environments would strengthen external validity.

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